

Sarasi Madahasi

Undergraduate | Computer Engineering | University of Ruhuna

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Summary

Third-year Computer Engineering undergraduate at the Faculty of Engineering, University of Ruhuna with a strong interest in Artificial Intelligence and Machine Learning. Experienced in developing machine learning applications in computer vision, natural language processing, and predictive analytics. Skilled in Python based ML development, data analysis, and model evaluation. Passionate about applying AI to solve real world problems in healthcare, safety, and intelligent systems

Education

BSc (Hons) in Computer Engineering

University of Ruhuna - Faculty of Engineering

2023 – Present

Hapugala, Galle, Sri Lanka

Related Coursework - Artificial Intelligence, Machine Learning, Image Processing & Computer Vision, Data Structures & Algorithms, Web Application Development, Operating Systems & Computer Networks, Database Systems, Object-Oriented Programming & Design Patterns, Software Architecture & Software Testing

GCE A/L 2021(2022) (First Attempt) - Physical Science

Sujatha Vidyalaya Matara

2019 – 2022 | Matara, Sri Lanka

Skills

Machine Learning & Data Science — Data Preprocessing, ANN, CNN, KNN, NLP, Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Model Evaluation

Programming Languages — C, C++, Python, Java, C#, JavaScript, Assembly (NASM)

Web & Application Development — HTML, CSS, JavaScript, React.js, Node.js, Express.js, Spring Boot (Java), Figma

DevOps and Cloud Computing — Linux administration, CI/CD, Docker, Jenkins, and AWS (Fundamentals)

Tools and Technologies — Git, Jupyter Notebook, Google Colab, Postman, Jira, SonarQube, AWS (Fundamentals), Docker, Jenkins, Linux

Soft Skills — Project Management, Critical Thinking, Problem-Solving, Communication, Leadership, and Teamwork

Projects

HealthForecast - AI-Powered Personal Health Risk Prediction Web App Using Daily Lifestyle Data (Ongoing)

- **Functionality:** Predicts potential health risks from user lifestyle data and generates personalized health recommendations using machine learning.
- **Technologies:** Python, Scikit-learn, Pandas, NumPy, Machine Learning.

Skin Cancer Stage Detection System

- **Functionalities:** Developed an AI-powered dermatology diagnostic system using U-Net++ for lesion segmentation, CNN for melanoma classification, ABCD feature extraction.
- **Technologies used:** Python, TensorFlow/PyTorch, U-Net++, CNN, OpenCV, ABCD feature extraction

Sentiment Analysis Web App

- **Functionalities:** Analyzes product reviews in real-time, classifies them as positive or negative, displays confidence scores, visualizes sentiment trends, and provides an interactive web interface.
- **Technologies used:** Python, Scikit-learn, Pandas, NumPy, NLTK, Logistic Regression, Flask, HTML, CSS, Data Visualization

Sleep Quality Prediction Using Lifestyle Factors

- **Functionality:** Develops a machine learning model to predict sleep quality based on lifestyle and health factors, comparing classification performance of different algorithms and analyzing the impact of lifestyle variables on sleep patterns.

- **Technologies used:** Python, Scikit-learn, Pandas, NumPy, Logistic Regression, KNN, Google Colab, Matplotlib.

VisionX - Smart Vision-Based Occupancy Detection & Energy Management System

(Ongoing Group Project)

- **Functionalities:** Real-time monitoring of indoor occupancy to optimize lighting, ventilation, and device control for energy savings using computer vision.
- **Technologies:** Python, OpenCV, TensorFlow/Keras, IoT devices, RGB/thermal cameras

Context-Aware Road Accident Risk Prediction Using ANN

(Ongoing Group Project)

- **Functionality:** Predicts low, medium, and high traffic accident risks based on contextual driving conditions to support proactive road safety.
- **Technologies:** Python, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, ANN, data preprocessing.

Pawish - Pet Care & Adoption Platform

- **Functionalities:** Developed a web-based pet care and adoption platform with user and admin dashboards for managing pet rescue listings, adoption applications, and volunteer donations.
- **Technologies used:** React.js, CSS, Node.js, Express.js and MongoDB.

More projects available at: sarasii22.github.io/portfolio 

COMPETITIONS & ACHIEVEMENTS

IEEE Xtreme 19.0 - Global Programming Competition (In Progress)

10th Place - Sri Lanka | 261st Place - Global

Devthon 2.0- Organized by the Leo Club of University of Moratuwa

Finalists (Top 12)

LICENSES AND CERTIFICATIONS

AI/ML Engineer - Stage 1

Centre for Open and Distance Education , Faculty of Computing, SLIIT

AI/ ML Engineer - Stage 2 (Ongoing)

Centre for Open and Distance Education , Faculty of Computing, SLIIT

Python for Beginners

Centre for Open & Distance Learning, University of Moratuwa

Introduction to Data Science

Cisco Networking Academy

Diploma in AI for Healthcare Professionals (Ongoing)

Alison

Volunteering Experience

IEEE - Xtream 19.0 - Student Branch University of Ruhuna

2025

Team Lead - Program Team

ReXtro - 2025 Sliver Jubilee Exhibition

2025

Team member - Event Management Committee
(Team Lead - Guest Lectures)

Mehewara

2023 – 2024

Mehewara to support students in rural schools preparing for their GCE O/L exams. Focused on teaching Mathematics, helping students strengthen problem-solving skills and exam readiness, while developing experience in mentoring and educational support.

References

Mr. P.A.D. Shehan Nilmantha Wijesekara, Lecturer,

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